

Uber Trip Data Analysis

Data Science with Python Project Report

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Repository: [SKNOMERCY/Uber-Trip-Data-Analysis](#)

1. Project Objective

Analyze Uber trip data using Python to understand travel patterns, demand trends, time-based behavior, and location patterns.

The project applies data loading, cleaning, exploratory data analysis, visualization, peak-hour analysis, and business interpretation to derive useful insights from trip records.

2. Dataset and Tools

The project uses an Uber trip dataset containing trip-level information such as start and stop locations, dates and times, category, purpose, distance, and related trip attributes.

Tools used: Python, Pandas, Matplotlib, Seaborn, and Jupyter Notebook.

3. Data Cleaning and Preprocessing

The notebook performs preprocessing needed for analysis, including handling missing values, removing duplicate records where applicable, converting date/time information into usable datetime values, and standardizing fields used in analysis.

Date/time features are then used to derive additional demand-related variables such as hour and day of week.

4. Exploratory Data Analysis

The analysis examines trip categories and purposes, monthly patterns, distance and duration, round-trip behavior, correlations, and start/stop locations.

Visualizations are used to make the distributions and patterns easier to interpret and to support the conclusions drawn from the data.

5. Peak Hours and Demand Trends

The completed notebook adds a dedicated peak-hours and demand-trends section. It derives HOUR and DAY_OF_WEEK from START_DATE and visualizes demand by hour, by day, and through a day-by-hour heatmap.

Observed peak hour: 3:00 PM (15:00), with 98 trips.

Observed busiest day: Friday, with 206 trips.

The heatmap provides a combined view of how trip activity varies across hours and days of the week.

6. Business Insights

The time-based analysis can help identify periods of relatively higher trip activity and therefore provides a basis for thinking about demand planning and resource allocation.

Location and purpose analysis can help reveal recurring travel patterns and the contexts in which trips occur.

These observations are descriptive of the analyzed dataset and should be interpreted as patterns in the sample rather than universal Uber demand behavior.

7. Conclusion

This project demonstrates an end-to-end workflow for Uber trip data analysis using Python. The workflow covers data preparation, exploratory analysis, visualization, time-based demand analysis, and interpretation.

The completed notebook provides reproducible analysis and visual evidence for the main findings, including the identified peak hour and busiest day.

8. Project Repository

GitHub repository: <https://github.com/SKNOMERCY/Uber-Trip-Data-Analysis>

Notebook: `notebook/Uber_Trip_Analytics.ipynb`

Key Results

Metric	Observed result
Peak hour	3:00 PM (15:00)
Trips at peak hour	98
Busiest day	Friday
Trips on busiest day	206

Note: Results above are based on the completed project notebook and should be interpreted in the context of the analyzed dataset.